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// Pin Definitions
#define TRIG_PIN 2
#define ECHO_PIN 3
#define RELAY_PIN 4
#define IR_SENSOR 5
#define GREEN_LED 6
#define RED_LED 7
long duration;
int distance;
void setup() {
pinMode(TRIG_PIN, OUTPUT);
pinMode(ECHO_PIN, INPUT);
pinMode(RELAY_PIN, OUTPUT);
pinMode(IR_SENSOR, INPUT);
pinMode(GREEN_LED, OUTPUT);
pinMode(RED_LED, OUTPUT);
digitalWrite(RELAY_PIN, LOW); // Lift OFF
Serial.begin(9600);
}
void loop() {
// Ultrasonic Sensor to detect car
digitalWrite(TRIG_PIN, LOW);
delayMicroseconds(2);
digitalWrite(TRIG_PIN, HIGH);
delayMicroseconds(10);
digitalWrite(TRIG_PIN, LOW);
duration = pulseIn(ECHO_PIN, HIGH);
distance = duration * 0.034 / 2;
Serial.print("Distance: ");
Serial.println(distance);
// Car detected
if (distance < 10) {
digitalWrite(GREEN_LED, HIGH);
digitalWrite(RED_LED, LOW);
// Activate hydraulic lift
digitalWrite(RELAY_PIN, HIGH);
delay(5000); // Lift goes UP
// Check parking slot using IR sensor
if (digitalRead(IR_SENSOR) == LOW) {
digitalWrite(RELAY_PIN, LOW); // Stop lift
Serial.println("Car Parked Successfully");
}
}
else {
digitalWrite(GREEN_LED, LOW);
digitalWrite(RED_LED, HIGH);
digitalWrite(RELAY_PIN, LOW);
}
delay(1000);
}
```